Solving Heterogeneous Agent Models with Nonconvex Optimization Problems: Linearization and Beyond

Michael Reiter, IHS, Vienna and NYU Abu Dhabi

November 13, 2018

Outline

Introduction

Linearization

State and Value Function Aggregation

Global approximations

More stuff

Technical problem of HA models

High dimensionality of the state space

- Impossible to solve agent optimization problem exactly.
- Impossible to follow exactly the dynamics of high-dimensional aggregate systems.

Example 1: Chang/Kim, AER 2007

- Heterogeneous households: idiosyncratic labor productivity follows Markov process
- Indivisible labor: work either zero hours or fixed number of hours
- Otherwise no labor market friction
- Objective: study observed ”labor wedge”: discrepancy between wage and observed MRS between consumption and leisure.
  Why does
  $$w = \frac{U_L}{U_C} = \eta \frac{C}{1-L} \tag{1}$$
  not hold over the business cycle?

Technology:
Production function:

$$Y_t = F(L_t, K_t, \lambda_t) = \lambda_t L_t^{\alpha} K_t^{1-\alpha} \tag{2}$$

where λ_t is Markov with transition probability distribution π_λ.

Chang/Kim 2007,Value function

$$V(a,x;\lambda,\phi) = \max_{a'\in\mathcal{A},h\in\{0,\bar{h}\}} \left\{ u(c,h) + \beta \, \mathrm{E} \, V(a',x';\lambda',\phi') \right\} \quad (3)$$

s.t.

$$c = w(\lambda,\phi)xh + (1 + r(\lambda,\phi))a) - a' \quad (4)$$

$$a' \geq \bar{a} \quad (5)$$

$$\phi' = T(\lambda,\phi) \quad (6)$$

where

- x is exogenous individual productivity process
- λ is aggregate TFP process
- ϕ is cross-sectional distribution of agents over (a,x).

CKResults

Properties of solution

Labor supply is determined by extensive margin: work or not

- Reaction of switch point to change in environment
- Fraction of households at the margin

Quality of linear approximation will depend on smoothness of cross-sectional distribution around switch points.

”Classical” Method: Krusell/Smith

- Agents assume aggregate law of motion:

$$\log K_{t+1} = \gamma_{K,0} + \gamma_{K,1} \log K_t + \gamma_{K,2}\lambda_t \tag{7a}$$
$$\log w_{t+1} = \gamma_{w,0} + \gamma_{w,1} \log K_t + \gamma_{w,2}\lambda_t \tag{7b}$$

- Agents optimize subject to (7).
- Equilibrium (fixed point): find γ that is OLS estimator of (7) in long model simulation.
- Accuracy check: R^2 of (7) in simulation.

My aim: avoid the assumption (7).

My approach

- Finite but high-dimensional approximation of model.
- Linearization around stationary state without aggregate shocks (Reiter 2009)
- Almost exact state and value function reduction of linearized model (Reiter 2010a).
- Global approximation method:
  - use value function from linearized model as starting point
  - backward iteration in time always solving for temporary equilibrium (Reiter 2010b)
  - concrete question: how useful are reduced states from linearized model for global approximiation?
    Ways to fined smaller/better reduced state space?
- Toolkit implemented in Julia

CKResults

Recent literature on HA methods

- Ahn, Kaplan, Moll, Winberry, and Wolf (2018): linearized model in continuous time, state reduction similar to Reiter (2010a); smooth approximation of value function
- Kubler and Scheidegger (2018): optimal low-dimensional approximation, ”self-justified equilibria”, ”bounded rationality”
- Winberry (2018): low-dimensional smooth approximation of cross-sectional distribution
- Boppart, Krusell, and Mitman (2018): Linearized solution by superposition of impulse responses (response to ”MIT” shock)
- Mertens and Judd (2017): perturbation around deterministic steady state (no shocks at all)
- Childers (2016): theoretical foundation of finite-dimensional approximations

Outline

Introduction

Linearization

State and Value Function Aggregation

Global approximations

More stuff

Simple Abstract Heterogeneous Agent Model

- Continuum (mass 1) of agents, ex ante identical.
  Index an agent by subscript h.
- Individual endogenous state: $x_{h,t} \in \{\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_N\}$.
- Individual exogenous shock: $z_{h,t} = \bar{z}_j$ with probability p_j, $j = 1, \ldots, J$.
  $z_{h,t}$ is i.i.d. across agents and time (so it is not a state, for ease of notation).
- Agents differ ex post because of individual history $z_{h,t}$.
- Aggregate exogenous process Z_t (first order Markov).

Agents' objective

$$\max_{a_{h,t}} \mathrm{E} \sum_{t=0}^{\infty} \beta^t U(a_{h,t}, x_{h,t-1}, z_{h,t}, Z_t, X_{t-1}, A_t) \tag{8}$$

subject to $a_{h,t} \in G(x_{h,t-1})$ and

$$prob(x_{h,t} = \bar{x}_j | x_{h,t-1} = \bar{x}_i) = P_{i,j}(a_{h,t}, x_{h,t-1}, z_{h,t}, Z_t, X_{t-1}, A_t) \tag{9}$$

where

- $\phi_{i,t}$ is mass of agents at state $\bar{x}_i$ at end of period t.
- $X_t \equiv \sum_{i=1}^{N} \bar{x}_i \phi_{i,t}$ is average state.
- $A_t \equiv \sum_{i=1}^{N} a(\bar{x}_i) \phi_{i,t}$ is average action.
- $G(x_{h,t-1})$: set of feasible actions $a_{h,t}$ conditional on $x_{h,t-1}$.

Notice: the distribution $\vec{\phi}$ directly affects utility (8) and the transition function (9) only through the aggregates X and A!

Structure of exact solution

- Aggregate State Space
  N aggregate state variables:
  1. $\phi_{i,t}$, $i = 1, \ldots, N$ with $N - 1$ degrees of freedom
  2. Z_t
- Individual decision function

$$a_{h,t} = \alpha(x_{h,t-1}, z_{h,t-1}, \vec{\phi}_{t-1}, Z_t) \tag{10}$$

Aggregate Law of Motion

Given

- the individual decision function $\alpha(.)$
- last period's state $\vec{\phi}_{t-1}$
- this period's realizations Z_t (realizations of $z_{h,t}$ do not matter because of law of large numbers)

we get

$$\phi_{m,t} = \sum_{n=1}^{N} \sum_{j=1}^{J} \phi_{n,t-1} \cdot p_j \cdot P_{n,m}(\alpha(\bar{x}_n, \bar{z}_j, \vec{\phi}_{t-1}, Z_t), \bar{x}_n, \bar{z}_j, Z_t, X_{t-1}, A_t) \tag{11}$$

Denote ALM by $\vec{\Phi}(\vec{\phi}_{t-1}, Z_t)$.

Aggregate Law of Motion, Matrix Form

Notice that, for given decision function $\alpha(\bar{x}_n, \bar{z}_j, \vec{\phi}_{t-1}, Z_t)$, (11) is linear in $\vec{\phi}_{t-1}$, so we can write the ALM in matrix form as

$$\vec{\phi}_t = \Pi_t \vec{\phi}_{t-1}$$

where the elements of the transition matrix Π are given by

$$\Pi_{m,n,t} = \sum_{j=1}^{J} p_j \cdot P_{n,m}(\alpha(\bar{x}_n, \bar{z}_j, \vec{\phi}_{t-1}, Z_t, A_t), \bar{x}_n, \bar{z}_j, Z_t, X_{t-1})$$

Recursive equilibrium

- Agents' optimization: Bellman equation

$$V(\bar{x}_n, \vec{\phi}_{t-1}, Z_t) = \max_a \Bigg\{ \sum_j p_j \Big[U\left(a, \bar{x}_n, \bar{z}_j, Z_t, X_{t-1}, A_t\right)$$

$$+\beta \sum_{n=1}^{N} P_{n,m}(a, \bar{x}_n, \bar{z}_j, Z_t, X_{t-1}, A_t)\, \mathrm{E}_t\, V\left(\bar{x}_m, \vec{\Phi}(\vec{\phi}_{t-1}, Z_t), Z_{t+1}\right) \Big] \Bigg\} \tag{12}$$

  which gives optimal decision $\alpha(\bar{x}_n, \vec{\phi}_{t-1}, Z_t)$.
- $\vec{\Phi}(\vec{\phi}_{t-1}, Z_t)$, satisfies (11) given decision rule $\alpha(x, \vec{\phi}, Z)$.
- z_i and Z follow exogenous processes

Notice: expectation E_t is over Z_{t+1}.

Equation System for Linearization

$$Z_t = \rho Z_{t-1} + \varepsilon_t \tag{13}$$

FOCs (at points with interior solution) and values:

$$0 = \sum_j p_j \Big[U_a \left(a_{n,t}, \bar{z}_j, Z_t, X_{t-1} \right) + \beta \sum_{m=1}^{N} \frac{\partial P_{n,m}(a_{n,t}, \bar{x}_n, \bar{z}_j, Z_t, X_{t-1})}{\partial a_{n,t}} \mathrm{E}_t \, V_{t+1}(\bar{x}_m) \Big] \tag{14}$$

$$V_t(\bar{x}_n) = \sum_j p_j \Big[U \left(a_{n,t}, \bar{z}_j, Z_t, X_{t-1} \right) + \beta \sum_{m=1}^{N} P_{n,m}(a_{n,t}, \bar{x}_n, \bar{z}_j, Z_t, X_{t-1}) \, \mathrm{E}_t \, V_{t+1}(\bar{x}_m) \Big] \tag{15}$$

Distribution dynamics (11) and definition

$$X_t \equiv \sum^{N} \bar{x}_i \phi_{i,t} \tag{16}$$

Equation System, comments

- If optimal choice is interior, differentiate, FOC. Otherwise, differentiate constraint.
- For linearization, it is convenient to assume that Z_t follows an AR(1) process.
- The dependence of decisions and values on the aggregate state $(\vec{\phi}_{t-1}, Z_t)$ is captured by the time subscript $a_{n,t}$, $V_t(\bar{x}_n)$.
- Individual and aggregate uncertainty are handled differently:
  1. Expectations over individual shocks are written explicitly as sums (or integrals) over realizations $(\sum_j p_j, \sum_{m=1}^{N} P_{n,m})$.
  2. Expectations over aggregate shocks is captured by the expectation operator E_t.

Equation System for Linearization, ctd.

- Equs. (11), (13), (14), (15) and (16) define a dynamic system of $3N + 2$ nonlinear equations in $3N + 2$ variables in each period t:
  - Z_t
  - $a_{n,t}, n = 1, \ldots, N$
  - $\phi_{n,t}, n = 1, \ldots, N$
  - X_t
  - $V_t(\bar{x}_n), n = 1, \ldots, N$

Notice: since $\sum_{i=1}^{N} \phi_i = 1$, one equation in (11) is redundant and will be replaced by $\sum_{i=1}^{N} \tilde{\phi}_i = 1$.

Steady state without aggregate shocks

- Set all Z_t (but not the $z_{h,t}$!) to their unconditional expectation Z^*.
- Then Equs. (11), (13), (14), (15) and (16) define a static system of $3N+1$ nonlinear equations in the $3N+2$ variables
  Z^* and $(a_n^*, \phi_n^*, V^*(\bar{x}_n))$ for $n = 1, \ldots, N$.
- Define

$$X^* \equiv \sum_{i=1}^{N} \bar{x}_i \phi_i^* \tag{17}$$

The solution can be computed by standard methods:

1. Guess X^*
2. Given X^*, compute the HH solution (by DP) and the corresponding distribution ϕ^*.
3. Iterate over X^* until (17) is satisfied.

Linearization around steady state (Reiter 2009)

- Stack all variables into the vector

$$\theta_t = \begin{bmatrix} Z_t \\ X_t \\ a_{n,t}, n = 1, \ldots, N \\ \phi_{n,t}, n = 1, \ldots, N \\ V_t(\bar{x}_n), n = 1, \ldots, N \end{bmatrix}$$

- The linearized model takes the form

$$\Lambda\theta_{t-1} + \Gamma\theta_t + \mathrm{E}_t\,\Phi\theta_{t+1} + \Psi\epsilon_t = 0 \tag{18}$$

- To obtain (18), differentiate the $3N + 2$ equations (11), (13), (14), (15), (16) w.r.t. all elements of θ_{t-1}, θ_t, θ_{t+1}, at the deterministic steady state.

Differentiating the value function

Take any variable $\Omega \in (\theta_{-1}, \theta_0, \theta_1)$. Differentiating (15) gives

$$\frac{\partial V_t(\bar{x}_n)}{\partial \Omega} = \sum_j p_j \Bigg[\frac{\partial U}{\partial \Omega} + U_a \frac{\partial a}{\partial \Omega} + \beta \sum_{m=1}^{N} \left(\frac{\partial P_{n,m}}{\partial \Omega} + \frac{\partial P_{n,m}}{\partial a} \frac{\partial a}{\partial \Omega} \right) \mathrm{E}_t \, V_{t+1}(\bar{x}_m) + P_{n,m} \frac{\partial V_{t+1}(\bar{x}_m)}{\partial \Omega} \Bigg] \tag{19}$$

Problem: $\frac{\partial a}{\partial \Omega}$ is not known, but

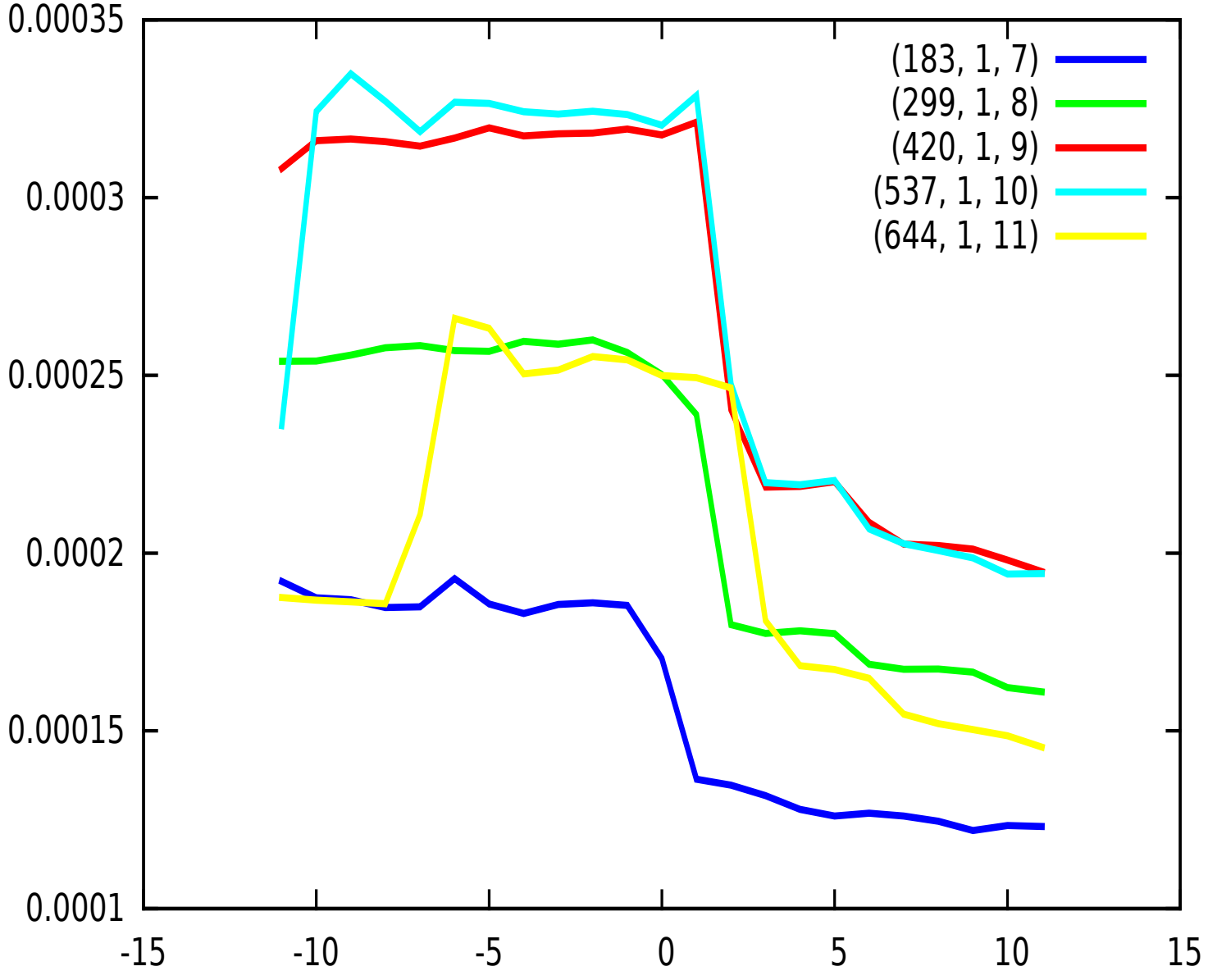
1. Interior solution: use envelope condition, $\frac{\partial a}{\partial \Omega}$ disappears
2. Boundary condition binds: differentiate boundary

Interpreting the linearized solution

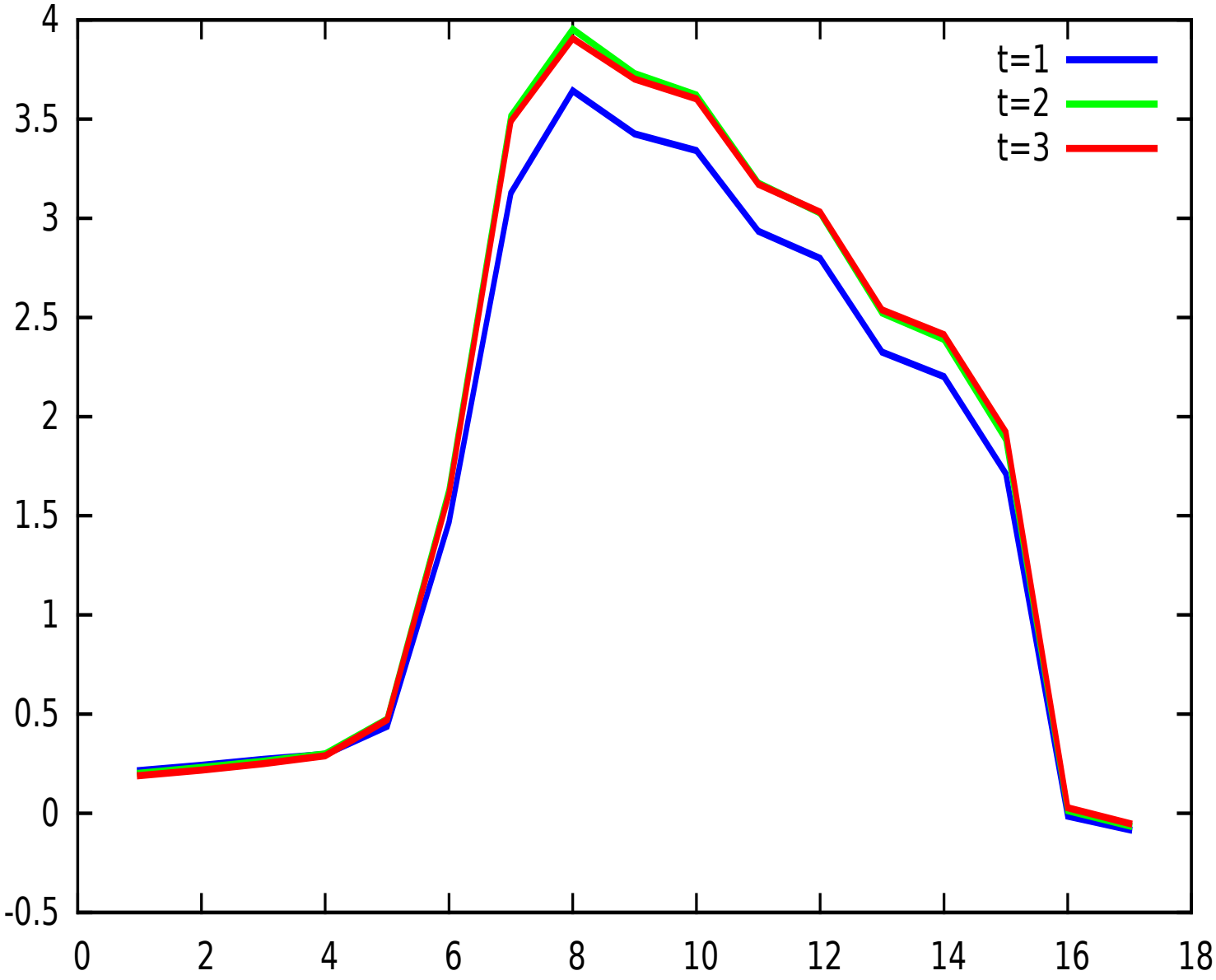
- The individual problem (decision as a function of the individual states) is solved nonlinearly by DP (or other techniques)
- The dependance of the individual solution on aggregate states (the cross-sectional distribution of assets etc.) is linearized.
- Precautionary behavior:
  - Precaution w.r.t. individual shocks is captured.
  - Precaution w.r.t. aggregate shocks is not captured (certainty-equivalence w.r.t. aggregate shocks).

CKResults

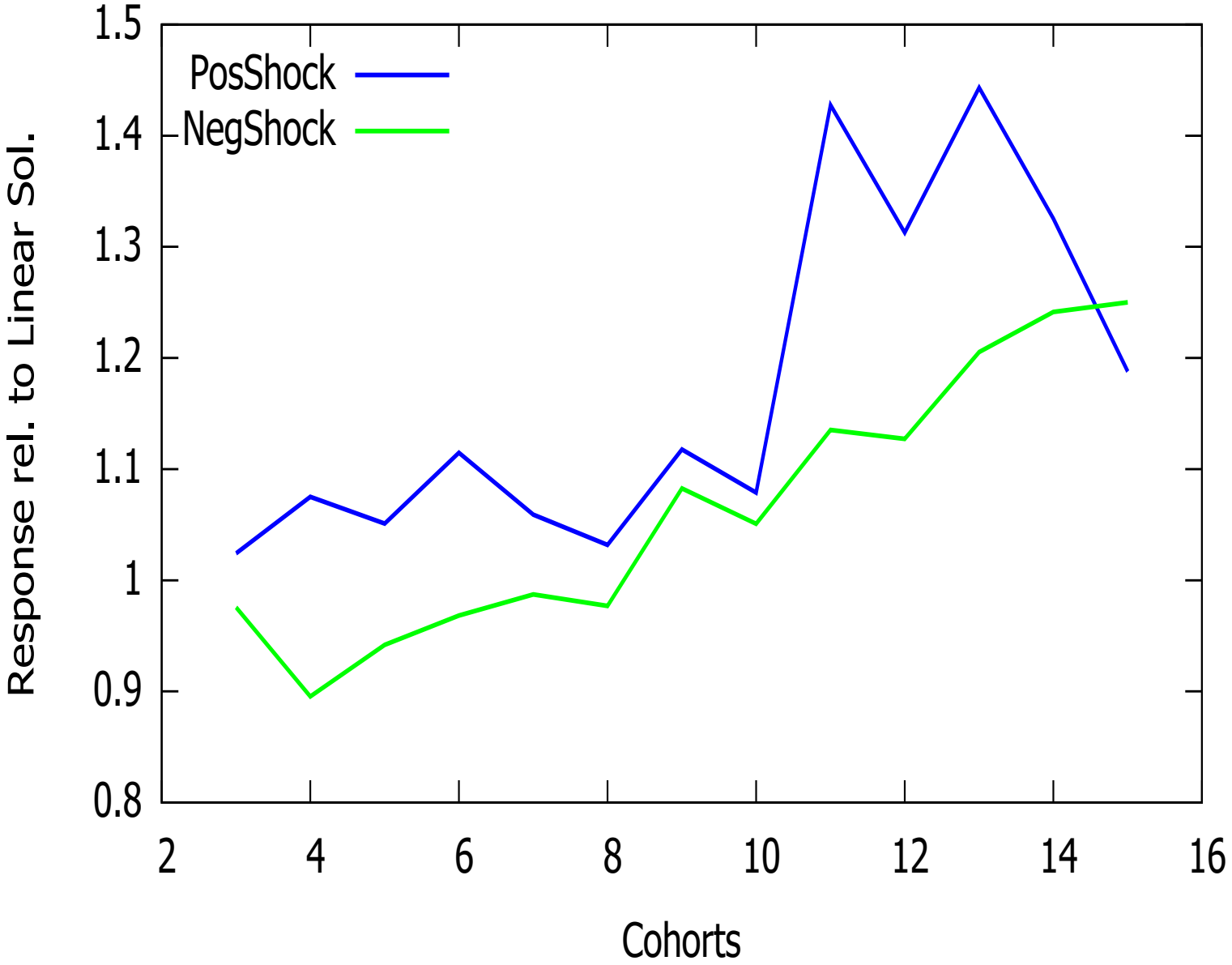
CK model, distribution around switch points



CK model, IR of switch points, linearized



CK model, IR of switch points, nonlinear



Outline

Introduction

Linearization

State and Value Function Aggregation

Global approximations

More stuff

Split equation system

Organize the linear equation system as

$$\begin{bmatrix} \Lambda_{ss} & 0 & 0 \\ \Lambda_{ys} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} s_{t-1} \\ y_{t-1} \\ v_{t-1} \end{bmatrix} + \begin{bmatrix} \Gamma_{ss} & \Gamma_{sy} & \Gamma_{sv} \\ \Gamma_{ys} & \Gamma_{yy} & \Gamma_{yv} \\ 0 & \Gamma_{vy} & \Gamma_{vv} \end{bmatrix} \begin{bmatrix} s_t \\ y_t \\ v_t \end{bmatrix} + \mathrm{E}_t \begin{bmatrix} 0 & 0 & \Phi_{sv} \\ 0 & 0 & \Phi_{yv} \\ 0 & 0 & \Phi_{vv} \end{bmatrix} \begin{bmatrix} s_{t+1} \\ y_{t+1} \\ v_{t+1} \end{bmatrix} + \begin{bmatrix} \Psi_s \\ \Psi_y \\ 0 \end{bmatrix} \epsilon_t = 0 \quad (20)$$

- Only v appears with time index $t+1$
- Only s appears with time index $t-1$
- Only y, not s enters equations for v.
- We assume Γss, Γyy, Γvv are regular.
- We assume Γss and Γvv are easy to invert.

How to split

- s are all the variables that appear with time index $t-1$.
- v are all the variables that appear with time index $t+1$.
- y are all the other variables.
- Split the equation systems into equations so as to satisfy the constraints implicit in (20). This cannot be fully automatized, therefore: name each state transition equation equal to the name of the state variable.

Normalized equation system

Rewrite (20) as

$$\begin{bmatrix} \Gamma_{ss}^{-1}\Lambda_{ss} \\ \Lambda_{ys} \\ 0 \end{bmatrix} s_{t-1} + \begin{bmatrix} I & \Gamma_{ss}^{-1}\Gamma_{sy} & \Gamma_{ss}^{-1}\Gamma_{sv} \\ \Gamma_{ys} & \Gamma_{yy} & \Gamma_{yv} \\ 0 & \Gamma_{vv}^{-1}\Gamma_{vy} & I \end{bmatrix} \begin{bmatrix} s_t \\ y_t \\ v_t \end{bmatrix} + \mathrm{E}_t \begin{bmatrix} \Gamma_{ss}^{-1}\Phi_{sv} \\ \Phi_{yv} \\ \Gamma_{vv}^{-1}\Phi_{vv} \end{bmatrix} v_{t+1} + \begin{bmatrix} \Gamma_{ss}^{-1}\Psi_s \\ \Psi_y \\ 0 \end{bmatrix} \epsilon_t = 0 \quad (21)$$

Model Reduction (Aggregation)

- State reduction: choose selection matrix $\bar{M}$ that computes statistics (”moments”) of the cross-sectional distribution

$$m_t = \bar{M} s_t \tag{22}$$

  $dimen(m) << dimen(s)$.
- Value reduction: choose matrix $\bar{V}$ with $\bar{V}' \bar{V} = I$ that spans the space in which the value function lies:

$$v_t = \bar{V} f_t \tag{23}$$

  $dimen(f) << dimen(v)$.

Almost Exact Value Function Reduction

Forward looking equation system (simplifed notation)

$$v_t = -\mathrm{E}_t\left(\Gamma_{vy} y_t + \Phi_{vv} v_{t+1}\right) \tag{24}$$

Iterating gives

$$\begin{aligned} v_t &= -\mathrm{E}_t\left[\Gamma_{vy} y_t + \Phi_{vv}\Gamma_{vy} y_{t+1} + \Phi_{vv}^{y}\Gamma_{vy} y_{t+y} + \ldots\right] \\ &= -\mathrm{E}_t \sum_{i=0}^{\infty} \Phi_{vv}^{i}\Gamma_{vy} y_{t+i} \end{aligned} \tag{25}$$

- v_t is spanned by the $\Phi_{vv}^{i}\Gamma_{vy}$.
- $\bar{V}$: orthonormal basis of $\cup_{i=0}^{k}\Phi_{vv}^{i}\Gamma_{vy}$ for some finite k
  Essential: Γ_{vy} has few colums.
- Value reduction: $\cup_{i=0}^{k}\Phi_{vv}^{i}\Gamma_{vy}$ has small numerical rank.
  (Chang/Kim: reduction from dimension 6800 to 253.)

Normalized and semi-reduced equation system

Inserting (22) and (23) into (21) we get

$$\begin{bmatrix} \bar{M}\Gamma_{ss}^{-1}\Lambda_{ss} \\ \tilde{\Lambda}_{ys}\bar{M} \\ 0 \end{bmatrix} s_{t-1} + \begin{bmatrix} I & \bar{M}\Gamma_{ss}^{-1}\Gamma_{sy} & \bar{M}\Gamma_{ss}^{-1}\Gamma_{sv}\bar{V} \\ \tilde{\Gamma}_{ys} & \Gamma_{yy} & \Gamma_{yv}\bar{V} \\ 0 & \bar{V}'\Gamma_{vv}^{-1}\Gamma_{vy} & I \end{bmatrix} \begin{bmatrix} m_t \\ y_t \\ f_t \end{bmatrix} +$$

$$\mathrm{E}_t \begin{bmatrix} \bar{M}\Gamma_{ss}^{-1}\Phi_{sv}\bar{V} \\ \Phi_{yv}\bar{V} \\ \bar{V}'\Gamma_{vv}^{-1}\Phi_{vv}\bar{V} \end{bmatrix} f_{t+1} + \begin{bmatrix} \bar{M}\Gamma_{ss}^{-1}\Psi_s \\ \Psi_y \\ 0 \end{bmatrix} \epsilon_t = 0 \quad (26)$$

State reduction (aggregation)

- Choose an $n_m \times n_s$ matrix $\bar{M}$ with $n_m < n_s$ and define

$$m_t = \bar{M} s_t \tag{27}$$

  Interpretation: m_t denotes the statistics (”m” is a memo of ”moments) of the cross-sectional distribution that agents based their decision on (bounded rationality). We assume that those statistics are linear functions of the distribution.

- The matrix should be such that there exist $\tilde{\Lambda}_{ys}$ and $\tilde{\Gamma}_{ys}$ with

$$\Lambda_{ys} = \tilde{\Lambda}_{ys}\bar{M}, \qquad \Gamma_{ys} = \tilde{\Gamma}_{ys}\bar{M} \qquad \Gamma_{vs} = \tilde{\Gamma}_{vs}\bar{M} \tag{28}$$

  (28) is satisfied if the rows of Λ_{ys} and Γ_{ys} are spanned by the rows of $\bar{M}$:

$$\begin{bmatrix} \Lambda'_{ys} & \Gamma'_{ys} & \Gamma'_{vs} \end{bmatrix} \in span(\bar{M}') \tag{29}$$

  Using (28) in the second block of equations in (20), the terms in s are replaced by terms in m.

State aggregation: proxy distributions

- Premultiplying the first block, of equations in (20) by $\bar{M}$, we obtain n_m rather than ns equations, which is right if we replace s_t by m_t.
- Problem: in the term $\bar{M}\Gamma_{ss}^{-1}\Lambda_{ss}s_{t-1}$, the s_{t-1} cannot be replaced by m_{t-1}.
- We therefore choose a matrix $\bar{P}$ which selects to any vector of statistics m_t one specific distribution (called "proxy distribution", (Reiter 2010b)) $s_t^{PD} = \bar{P}m_t$ which has those statistics. Therefore we require

$$\bar{M}\bar{P} = I_{n_m} \tag{30}$$

- Interpretation: the state equations in (20) are satisfied not at all possible state vectors s, but only at those that are "typical" distributions in the sense $s = \bar{P}m$ for any m.

State aggregation: smooth example

Choose $\bar{M}$ such that it selects adjacent bins:

$$\bar{M} = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 & \dots & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & \dots & 0 & 0 & 0 \\ & \dots & & & \dots & & \dots & \dots & & \\ 0 & 0 & 0 & 0 & 0 & 0 & \dots & 1 & 1 & 1 \end{pmatrix}$$

[Does not satisfy (28)!]
Then we can set

$$\bar{P} = \begin{pmatrix} 1/3 & 1/3 & 1/3 & 0 & 0 & 0 & \dots & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 1/3 & 1/3 & \dots & 0 & 0 & 0 \\ & \dots & & & \dots & & \dots & \dots & & \\ 0 & 0 & 0 & 0 & 0 & 0 & \dots & 1/3 & 1/3 & 1/3 \end{pmatrix}$$

Proxy distribution assumes equal ditribution within bins.
Alternatively, assume proxy distribution is proportional to steady state distribution within any bin.

Almost exact aggregation: Conditional Expectations Approach (Reiter 2010a)

- We want to predict y_{t+i} for $i = 0, \ldots, \infty$.
- Assume

$$\begin{aligned} y_t &= Cs_t \\ \mathrm{E}_t\, s_{t+i} &= A^i s_t \end{aligned} \tag{31}$$

- We need the following linear combinations of s:

$$Cs_t, \qquad CAs_t, \qquad CA^2 s_t, \qquad \ldots \tag{32}$$

- Stack them into

$$Q = \begin{bmatrix} C \\ CA \\ CA^2 \\ \ldots \\ CA^{n-1} \end{bmatrix} \tag{33}$$

Rank of Q?

Computing exact aggregation

Consider an $n \times n$ matrix A and an $m \times n$ matrix C and compute the $m \cdot n \times n$ observability matrix

$$Q = \begin{bmatrix} C \\ CA \\ CA^2 \\ \dots \\ CA^{n-1} \end{bmatrix} \tag{34}$$

Define $k \leq n$ as the rank of Q. The SVD of Q can be written as

$$Q = \begin{bmatrix} U_1 & U_2 \end{bmatrix} \begin{bmatrix} S & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} V_1' \\ V_2' \end{bmatrix} = U_1 S V_1' \tag{35}$$

$$S \equiv diag(\sigma_1, \dots, \sigma_k) \tag{36}$$

where U_1 and V_1 have dimension $m \cdot n \times k$, $U_1' U_1 = V_1' V_1 = I_k$.

Computing $\bar{M}$

From the Cayley-Hamilton theorem, there exists a Λ such that

$$QA = \Lambda Q \tag{37}$$

Using (35) in (37) we get $U_1 S V_1' A = \Lambda U_1 S V_1'$. Premultiplying by $S^{-1} U_1'$ we get $V_1' A = S^{-1} U_1' \Lambda U_1 S V_1'$ Set $\bar{M} = V_1'$ and $\hat{A} = \bar{M} V_1 \bar{M}'$, then

1. $\bar{M}\bar{M}' = I_k$
2. $\bar{M}$ can be interchanged with A:

$$\bar{M}A = \hat{A}\bar{M} \tag{38}$$

   with $\hat{A} = S^{-1} U_1' \Lambda U_1 S$ being a $k \times k$-matrix. From 1. it follows that $\hat{A} = \bar{M}A\bar{M}'$.
3. We have $C' \in span(\bar{M}')$:

$$C = \begin{bmatrix} I & 0 & \dots & 0 \end{bmatrix} Q = \left(\begin{bmatrix} I & 0 & \dots & 0 \end{bmatrix} U_1 S \right) \bar{M} \tag{39}$$

Almost exact state aggregation (Reiter 2010a)

Assume that Q has rank $k << n$ when choosing $A = \Gamma_{ss}^{-1}\Lambda_{ss}$ and C such that

$$\begin{bmatrix}\Lambda'_{ys} & \Gamma'_{ys} & \Gamma'_{vs}\end{bmatrix} \in span(C) \tag{40}$$

Then there is a $k \times n$-matrix $\bar{M}$ such that

- $\Lambda_{ys} = \tilde{\Lambda}_{ys}\bar{M}, \qquad \Gamma_{ys} = \tilde{\Gamma}_{ys}\bar{M} \qquad \Gamma_{vs} = \tilde{\Gamma}_{vs}\bar{M}$
- There exists an $n_m \times n_m$-matrix $\hat{A}$ such that

$$\bar{M}\Gamma_{ss}^{-1}\Lambda_{ss} = \hat{A}\bar{M} \tag{41}$$

Then $\bar{M}\Gamma_{ss}^{-1}\Lambda_{ss}s_t = \hat{A}m_t$.
Chang/Kim: reduction from dimension 1700 to 393.

Normalized and reduced equation system

Combining almost exact state and value reduction we have

1. $v_t = \bar{V}' f_t$ with $\bar{V}'\bar{V} = I$
2. $\Lambda_{ys} = \tilde{\Lambda}_{ys}\bar{M}, \qquad \Gamma_{ys} = \tilde{\Gamma}_{ys}\bar{M}, \qquad \Gamma_{vs} = \tilde{\Gamma}_{vs}\bar{M}$
3. $\bar{M}\Gamma_{ss}^{-1}\Lambda_{ss}s_t = \hat{A}m_t$

We can transform (21) to

$$\begin{bmatrix} \hat{A} \\ \tilde{\Lambda}_{ys} \\ 0 \end{bmatrix} m_{t-1} + \begin{bmatrix} I & \bar{M}\Gamma_{ss}^{-1}\Gamma_{sy} & \bar{M}\Gamma_{ss}^{-1}\Gamma_{sv}\bar{V} \\ \tilde{\Gamma}_{ys} & \Gamma_{yy} & \Gamma_{yv}\bar{V} \\ 0 & \Gamma_{vv}^{-1}\Gamma_{vy} & I \end{bmatrix} \begin{bmatrix} m_t \\ y_t \\ f_t \end{bmatrix} +$$

$$\mathrm{E}_t \begin{bmatrix} \bar{M}\Gamma_{ss}^{-1}\Phi_{sv}\bar{V} \\ \Phi_{yv}\bar{V} \\ \Gamma_{vv}^{-1}\Phi_{vv}\bar{V} \end{bmatrix} f_{t+1} + \begin{bmatrix} \bar{M}\Gamma_{ss}^{-1}\Psi_s \\ \Psi_y \\ 0 \end{bmatrix} \epsilon_t = 0 \quad (42)$$

Asymmetry state reduction versus value reduction

1. Iterating the Bellman equation forward we can compute a basis for the subspace in which the value function lives.
2. The subspace in which the cross-sectional distribution lives depends on the shocks. It cannot be determined without solving the model.
   The commutation property (41) says that, knowing the statistics m_t, the exact distribution does not matter for the solution. It does not give a good proxy distribution.
   Notice that the optimal state reduction does not depend on the shocks!

State reduction Chang/Kim model

Linearized model:

- Number of states reduced from 17000 to around 393
- Number of value function parameters reduced from 6800 to around 253
- Accuracy: aggregation error $\approx 10^{-12}$.

Reducing even further: Balanced Reduction

Define the matrices R, $\mathcal{P}$, Q, $\mathcal{Q}$, U, S, V, $\tilde{H}$ as follows:

$$RR' = \mathcal{P} \equiv \mathcal{L}\left(A, B, \Sigma_\epsilon\right) \tag{43}$$

$$QQ' = \mathcal{Q} \equiv \mathcal{L}\left(A', C'\right) \tag{44}$$

$$USV' = R'Q \tag{45}$$

$$\tilde{H} = S^{-1/2} V' Q' \tag{46}$$

Take S as the square matrix containing only the non-zero singular values (and drop the columns of U and rows of V corresponding to the zero singular values), so that S is invertible by construction.

Balanced Reduction, ctd.

With the variable transformation $\hat{x} = \tilde{H}x$, $\hat{A} = \tilde{H}A\tilde{H}^{-1}$, $\hat{B} = \tilde{H}B$, $\hat{C} = C\tilde{H}^{-1}$. we get

$$\mathcal{L}\left(\hat{A}, \hat{B}, \Sigma_\epsilon\right) = \tilde{H}\mathcal{L}\left(A, B, \Sigma_\epsilon\right)\tilde{H}' = \mathcal{L}\left(\hat{A}', \hat{C}'\right) = (\tilde{H}')^{-1}\mathcal{L}\left(A', C'\right)\tilde{H}^{-1} = \tag{47}$$

In the new vector $\hat{x}$ the variables are ordered such that $\hat{x}_i$ has both the i-th highest variance, and makes the i-th highest contribution to future values of y.
We pick the first k components of $\hat{x}$, or the first k rows of $\tilde{H}$, such that the diagonal elements $S_{i,i}$ are negligible for $i > k$:

$$H = \tilde{H}_{1:k,:} \tag{48}$$

Properties of balanced reduction

Performance guarantee (Antoulas 2005, Theorem 7.10):

$$distance(ExactModel, ReducedModel) \leq 2(\sigma_{k+1} + \ldots + \sigma_n)$$

- σ's: singular values in (45) that were omitted in the construction of H in (48).
- distance is the Hankel norm: maximum of response

$$\sqrt{\sum_{i=0}^{\infty} ||y_{t+i} - \hat{y}_{t+i}||^2}$$

to any sequence of past shocks ϵ_{t-i} of unit L^2 size:

$$\%label eq : HankelIn \sum_{i=0}^{\infty} ||\epsilon_{t-i}||^2 = 1$$

Outline

Introduction

Linearization

State and Value Function Aggregation

Global approximations

More stuff

Structure of HA models HetSol solves

- The agent problem has three state variables:
  1. One endogenous continuous state variable.
  2. An endogenous discrete state.
  3. An exogenous discrete state.
- The agent problem has two decision variables:
  1. Endogenous continuous choice variable: end-of-period continuous state.
  2. Endogenous discrete choice.
- Individual uncertainty: change of discrete states between end of period and beginning of next period. Transition probabilities can depend on aggregate state.

Approximating the value function

- Split Ω into predetermined component $\mathbf{D}$ (includes distribution) and current states Z (such as driving shocks).
- Approximate value function by a function of reduced state vector M rather than Ω:

$$V(k, e, z, \Omega) \approx \hat{V}(k, e, z, M) \tag{49}$$

where "moments" M are given by

$$M = H \begin{bmatrix} \mathbf{D} \\ Z \end{bmatrix} = H_D \mathbf{D} + H_Z Z \tag{50}$$

for some given matrix H.

- Approximate as a linear combination of basis functions:

$$\hat{V}(k, e, z, M) = \mathcal{P}(M; \gamma(k, e, z)) \equiv \sum_{i=1}^{n_p} \phi_i(M) \gamma_i(k, e, z)$$

with separate coefficients γ_i for each individual state k, e, z.

Expected continuation value

$\mathrm{E}_\Omega\, V'$ can be written as

$$\mathrm{E}_\Omega\, V' = \sum_{z'} \pi(z, z'; \Omega)\, \mathrm{E}_{Z'}\, \hat{V}\left(k', e', z', \mathcal{P}\left(H_D \mathbf{D}' + H_Z Z'; \gamma(k', e', z')\right)\right)$$

With finite approximation for LoM of aggegate Z:

$$\mathrm{E}_\Omega\, V' = \sum_{z'} \pi(z, z'; \Omega) \sum_{Z'} \pi_Z(Z, Z') \times \hat{V}\left(k', e', z', \mathcal{P}\left(H_D \mathbf{D}' + H_Z Z'; \gamma(k', e', z')\right)\right) \quad (51)$$

Since $\mathbf{D}'$ is predetermined, we only need to know the finite number of statistics $H_D \mathbf{D}'$ in order to compute the expected continuation value.

Temporary Equilibrium

Assume any $\hat{V}(k, e, z, M)$ given. A temporary equilibrium $(p, m', \mathcal{A})$ at any aggregate state $\mathbf{D}, Z$ consists of

- a set of equilibrium values p
- a set of end-of-period statistics m'
- a policy function $\mathcal{A}$

such that

- policy functions are optimal

$$[d(k, e, z), a(k, e, z)] = \underset{d,a}{\operatorname{argmax}}\, U(k, e, z, d, a; p, \Omega) + \beta \sum_{z'} \pi(z, z'; \Omega)\, \mathrm{E}_{\Omega'} \left\{ \hat{V}\left(k', e', z', m' + H_Z Z'\right) \right\} \quad (52)$$

- prices satisfy the equilibrium condtions

$$\mathcal{E}(p, \mathbf{D}, Z, \mathcal{A}) = 0 \quad (53)$$

- end-of-period statistics satisfy the transition law

$$m' = H_D \mathcal{T}(\mathbf{D}, Z, p, \mathcal{A}) \quad (54)$$

Four different solutions

1. **Lin** : Linearized solution
2. **NonL1** : Temporary equilibrium with expected continuation value from linearized solution. This shows the effect of the distribution on switch points, and captures nonlinear extensive margin effect.
3. **NonL2** : Temporary equilibrium with global linear value function. This does not yet incorporate precautionary effects w.r.t. aggregate variables.
4. **NonL3** : Temporary equilibrium with nonlinear value function

CKResults

Backward iteration algorithm, setup

1. Select a set of aggregate state variables.
2. Set up a finite grid in state aggregate state variables.
3. Initialize the value function $V_{T+1}\left(\bar{x}_i; \bar{\Omega}_j\right)$ for $i = 1, \ldots, n_k n_e n_z$ and $j = 1, \ldots, n_A$ at all points of the aggregate grid. These values are taken from the linearized solution.

Iterating backward

For $t = T : -1 : 1$ do

1. Compute the coefficients of the polynomial approximation separately for each $k, e, zj \in G_x$ by linear projection.
2. For each aggregate grid point M_j find (p, m') s.t.

$$\tilde{V}(k_i, e_i, z_i) = \sum_{z'} \pi(z_i, z'; \Omega)$$

$$\sum_{Z'} \pi_Z(Z, Z') \sum_{j=1}^{n_p} \phi_i\left(m'\right) \gamma_{j,t+1}\left(k_i, e_i, z'\right))$$

$$[d(x_i), a(x_i)] =$$

$$\underset{d,a}{\operatorname{argmax}} \left\{ U\left(x_i, d, a; p, M_j\right) + \beta \tilde{V}^{\mathcal{I}}\left(a, T(e_i, d), z_i\right) \right\} \quad (55)$$

where x_i stands for k_i, e_i, z_i.

3. Update the value function for all $i = 1, \ldots, n_k n_e n_z$

Backward iteration algorithm, fixed point

- Stop iterations if
- After convergence, the value function should approximately satisfy the recursive relationship

$$\begin{aligned} V_t(x_i, M_j) &= U\left(x_i, d(x_i, M_j), a(x_i, M_j)), p(M_j)\right) \\ &\quad + \beta \tilde{V}^{\mathcal{I}}\left(a(x_i, M_j), \left(., T(e_i, d(x_i, M_j)), z_i; m'(M_j)\right)\right) \end{aligned} \tag{56}$$

Results Chang/Kim Model

	C/K	Takahashi	Lin	LinV	GlobLin	GLobNL
σ_H	0.76	0.57	0.56	0.56	0.48	0.46
σ_{wedge}	0.76	0.24	0.22	0.22	0.18	0.17
σ_Y	1.28	1.30	1.24	1.25	1.18	1.18
σ_C	0.39	0.33	0.33	0.33	0.40	0.42
σ_I	3.06	3.08	3.13	3.14	2.84	2.78
σ_L	0.50	0.41	0.41	0.42	0.35	0.35
$\sigma_{Y/H}$	0.50	0.49	0.49	0.48	0.55	0.56
$\rho(H, wedge)$	0.87	0.95	0.95	0.97	0.96	0.97

Table: Results CK model

Chang/Kim Model: value function regressions

Regressors	Linear approx. $\frac{var(err)}{var(V_{t,i})}$)	Linear approx. $\frac{var(err)}{var(V_{t,i}-V_{t-1,i})}$)	Quadr. approx. $\frac{var(err)}{var(V_{t,i}-V_{t-1,i})}$)
Global Linear Solution in $K_t, \lambda_t, \lambda_{t-1}$			
K_t, λ_t	7.98e-04	1.13e-01	5.81e-02
$K_t, \lambda_t, \lambda_{t-1}$	6.04e-04	8.48e-02	2.75e-02
$K_t, \lambda_{t-1}, 4BalMom$	2.69e-05	3.59e-03	1.85e-03
$K_t, \lambda_{t-1}, 6BalMom$	7.55e-06	8.98e-04	5.53e-05
$K_t, \lambda_{t-1}, 10BalMom$	6.12e-06	7.24e-04	3.10e-05
Global Quadr. Solution in $K_t, \lambda_t, \lambda_{t-1}$			
K_t, λ_t	1.00e-03	9.21e-02	3.29e-02
$K_t, \lambda_t, \lambda_{t-1}$	8.34e-04	6.92e-02	7.87e-03
$K_t, \lambda_{t-1}, 4BalMom$	3.52e-04	3.19e-02	5.02e-03
$K_t, \lambda_{t-1}, 6BalMom$	9.91e-05	5.49e-03	1.61e-03
$K_t, \lambda_{t-1}, 10BalMom$	7.51e-05	4.54e-03	1.17e-03

Table: Projection errors, mean over V_i, adjusted for DF

Lumpy investment: value function regressions

Regressors	Linear approx. $\frac{var(err)}{var(V_{t,i})}$)	Linear approx. $\frac{var(err)}{var(V_{t,i}-V_{t-1,i})}$)	Quadr. approx. $\frac{var(err)}{var(V_{t,i}-V_{t-1,i})}$)
Global Linear Solution in $K_t, \lambda_t, \lambda_{t-1}$			
K_t, λ_t	2.42e-03	4.95e-02	4.67e-02
$K_t, \lambda_t, \lambda_{t-1}$	1.44e-03	2.92e-02	2.66e-02
$K_t, \lambda_{t-1}, 4BalMom$	3.21e-04	6.18e-03	2.43e-03
$K_t, \lambda_{t-1}, 6BalMom$	1.65e-04	3.02e-03	4.60e-04
$K_t, \lambda_{t-1}, 10BalMom$	1.04e-04	1.79e-03	1.58e-04

Table: Projection errors, mean over V_i, adjusted for DF

Outline

Introduction

Linearization

State and Value Function Aggregation

Global approximations

More stuff

Iterative Solution of Linearized Model

We can write the linear model in the form

$$\Lambda x_{t-1} + \Gamma x_t + \mathrm{E}_t \, \Phi x_{t+1} + \Psi \epsilon_t = 0 \tag{57}$$

Linearity implies that the expected value can be written

$$\mathrm{E}_t \, x_{t+1} = F x_t \tag{58}$$

for a (unknown) matrix F. This implies

$$x_t = -\left(\Gamma + \Phi F\right)^{-1} \left(\Lambda x_{t-1} + \Psi \epsilon_t\right) \tag{59}$$

Consistency with (58) requires

$$F = -\left(\Gamma + \Phi F\right)^{-1} \Lambda \tag{60}$$

Algorithm:

1. Initialize $F = 0$.
2. Iterate until convergence (Rendahl 2017; Higham 2002):

$$F \leftarrow \left(\Gamma + \Phi F\right)^{-1} \Lambda \tag{61}$$

Efficient computation of iterative solution

Split the vector of variables x_t in a vector of states s_t, a vector of static equilibrium variables y_t, and a vector of forward looking variables (appear with index $t+1$) v_t:

$$x_t = \begin{bmatrix} s_t \\ y_t \\ v_t \end{bmatrix} \tag{62}$$

Denote the length of these vectors by n_s, n_y and n_v, respectively.

$$\Phi F = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \Phi_2 \end{bmatrix} \begin{bmatrix} F_{11} & 0 & 0 \\ F_{21} & 0 & 0 \\ F_{31} & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 \\ \Phi_2 \end{bmatrix} \begin{bmatrix} F_{31} & 0 & 0 \end{bmatrix} \tag{63}$$

Efficient computation of iterative solution, ctd.

Use Sherman-Morrison-Woodbury formula:

$$\left(\Gamma + UV'\right)^{-1} = \Gamma^{-1} - \Gamma^{-1}U\left(I + V'\Gamma^{-1}U\right)^{-1}V'\Gamma^{-1} \tag{64}$$

- Case 1: few forward looking variables: $(\#rows(F_{31}) < \#cols(F_{31)})$

$$U = \begin{bmatrix} 0 \\ \Phi_2 \end{bmatrix}$$
$$V' = \begin{bmatrix} F_{31} & 0 & 0 \end{bmatrix}$$

  Dimension of $V'\Gamma^{-1}U$ is n_v.
- Case 2: few states $(\#rows(F_{31}) > \#cols(F_{31)})$

$$U = \begin{bmatrix} 0 \\ \Phi_2 F_{31} \end{bmatrix}$$
$$V' = \begin{bmatrix} I & 0 & 0 \end{bmatrix}$$

  Dimension of $V'\Gamma^{-1}U$ is n_s.

Example 2: RBC Model with Lumpy Investment

The model is similar to Thomas (2002).

- Aggregate shock: TFP (λ)
- Representative household
- Perfect competition on goods and labor markets
- Heterogeneous firms:
  - Decreasing returns to scale
  - Idiosyncratic productivity shocks
  - Nonconvex adjustment costs
  - All firms owned by representative households Stochastic discount factor

$$Q_{t,t+i} = \frac{U_c(C_{t+i}, L_{t+i})}{U_c(C_t, L_t)} \tag{65}$$

Lumpy Investment, Firm Problem

$$\max_{K_1,K_2,\ldots;L_1,L_2,\ldots} \mathrm{E} \sum_{t=1}^{\infty} Q_{0,t} D_t$$

- dividends $D_t = \lambda_t x_t K_{t-1}^{\alpha_K} L_t^{\alpha_L} - w_t L_t - \Phi(K_{t-1}, K_t)$
- λ is an exogenous Markov process (aggregate)
- x is an exogenous Markov process (idiosyncratic)
- Investment costs are

$$\Phi(K_{t-1}, K_t) = \begin{cases} \zeta K_{t-1} & \text{if } \Delta K_t = 0 \\ \zeta K_{t-1} + f + \Delta K_t & \text{if } \Delta K_t > 0 \\ \zeta K_{t-1} + f + \phi \Delta K_t & \text{if } \Delta K_t < 0 \end{cases} \tag{66}$$

  where $\Delta K_t \equiv K_t - K_{t-1}(1-\delta)$
- maintainance costs $\zeta \geq 0$, fixed costs $f \geq 0$
- revenues from selling capital $\phi \leq 1$

Ahn, S., G. Kaplan, B. Moll, T. Winberry, and C. Wolf (2018).
When Inequality Matters for Macro and Macro Matters for Inequality.
NBER Macroeconomics Annual 32(1), 1–75.

Antoulas, A. C. (2005).
Approximation of Large-Scale Dynamical Systems.
SIAM.

Boppart, T., P. Krusell, and K. Mitman (2018).
Exploiting MIT shocks in heterogeneous-agent economies: the impulse response as a numerical derivative.
Journal of Economic Dynamics and Control 89(C), 68–92.

Childers, D. (2016).
On the Solution and Application of Rational Expectations Models with Function-Valued States.
Technical report.

Higham, N. (2002).
Accuracy and Stability of Numerical Algorithms (2nd ed.).

SIAM.

Kubler, F. and S. Scheidegger (2018).
Self-justified equilibria: Existence and computation.
Manuscript, University of Zurich.

Mertens, T. M. and K. L. Judd (2017).
Solving an incomplete markets model with a large cross-section of agents.
?

Reiter, M. (2009).
Solving heterogenous agent models by projection and perturbation.
Journal of Economic Dynamics and Control 33(3), 649–665.

Reiter, M. (2010a).
Approximate and almost-exact aggregation in dynamic stochastic heterogeneous-agent models.
IHS Working Paper 258.

Reiter, M. (2010b).

Solving the incomplete markets model with aggregate uncertainty by backward induction.
Journal of Economic Dynamics and Control 34(1), 28–35.

Rendahl, H. (2017).
Linear time iteration.

Thomas, J. K. (2002).
Is lumpy investment relevant for the business cycle?
Journal of Political Economy 110(3), 508–534.

Winberry, T. (2018).
A toolbox for solving and estimating heterogeneous agent macro models.
Quantitative Economics, forthcoming.